

Heaven's Light is Our Guide



Rajshahi University of Engineering and Technology

Department of Electrical and Computer Engineering

Report 01

Course Title : Analog Electronic Circuits 2- Sessional

Course No: ECE 2112

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Experiment No-01

Problem Definition

Design and implement an **AND gate using only NAND gates**. Verify the circuit by comparing the output with the corresponding truth table.

Circuit Diagram

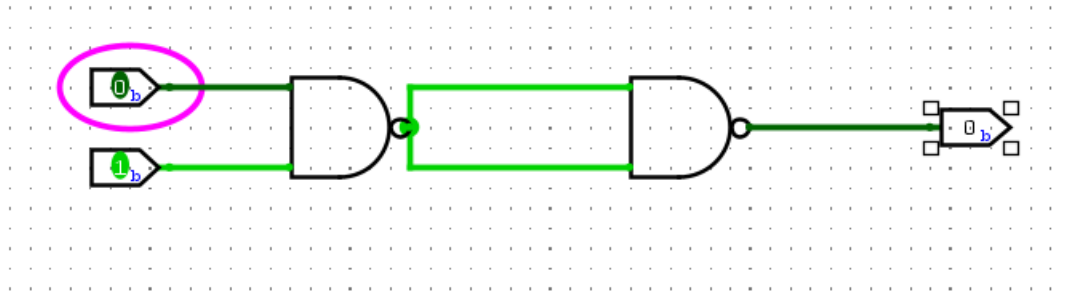


Figure 1: Logic Diagram of AND Gate using NAND Gates

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Expression

$$Y = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$$

Experiment No-02

Problem Definition

Design and implement an **OR gate using only NAND gates**. Verify the output using the truth table.

Circuit Diagram

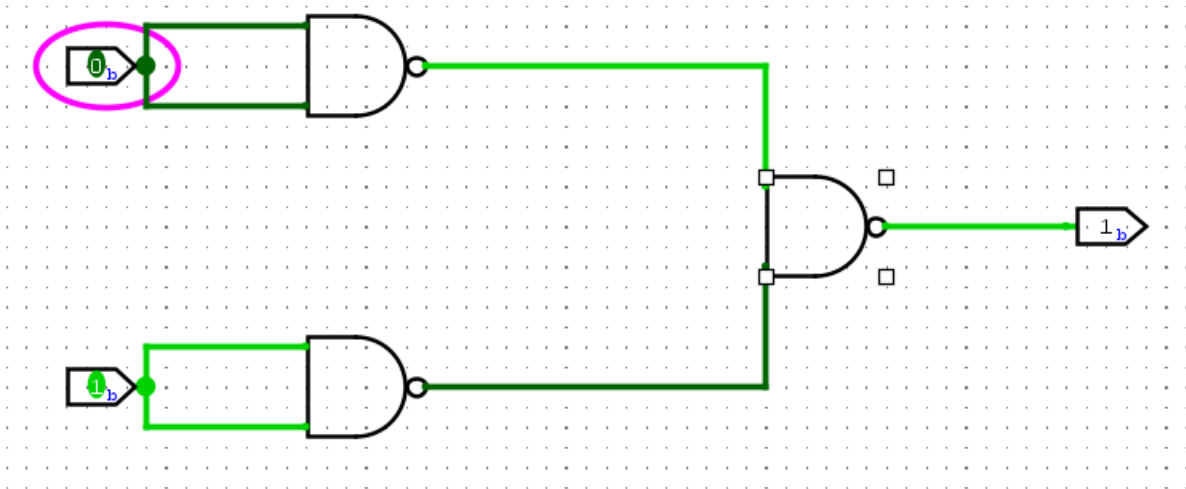


Figure 2: Logic Diagram of OR Gate using NAND Gates

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Expression

$$Y = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$$

Experiment No-03

Problem Definition

Design and implement a **NOT gate using only NAND gates**. Verify its operation using the truth table.

Circuit Diagram

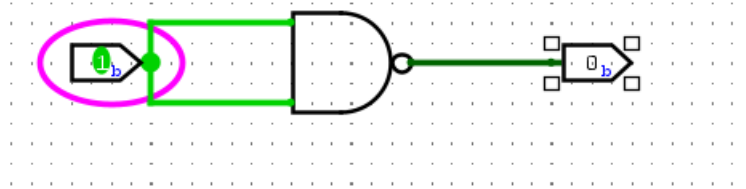


Figure 3: Logic Diagram of NOT Gate using NAND Gate

Truth Table

A	Y
0	1
1	0

Expression

$$Y = A \text{ NAND } A$$

Experiment No-04

Problem Definition

Design and implement an **AND gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram

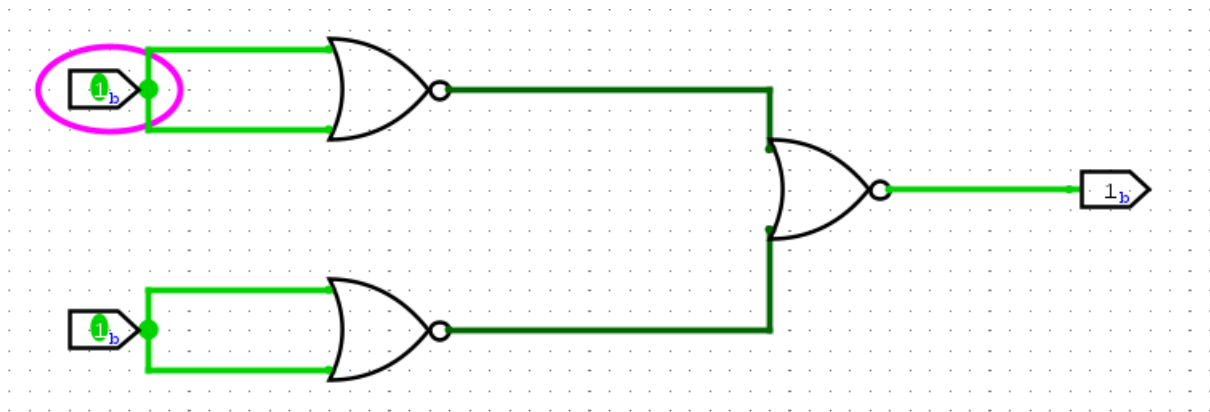


Figure 4: Logic Diagram of AND Gate using NOR Gates

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Expression

$$Y = (A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$$

Experiment No-05

Problem Definition

Design and implement an **OR gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram

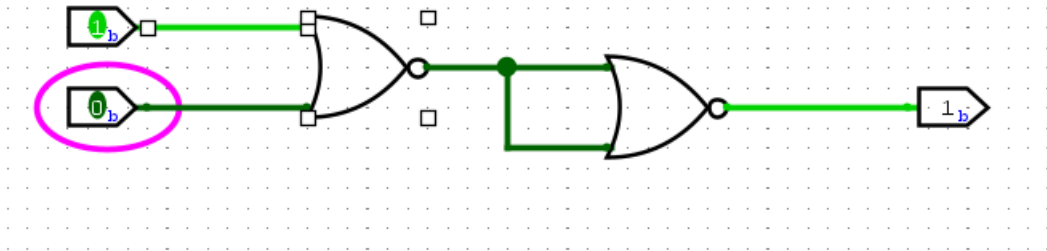


Figure 5: Logic Diagram of OR Gate using NOR Gates

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Expression

$$Y = (A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$$

Experiment No-06

Problem Definition

Design and implement a **NOT gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram

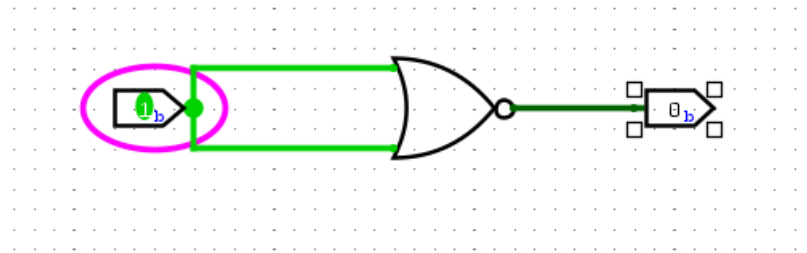


Figure 6: Logic Diagram of NOT Gate using NOR Gate

Truth Table

A	Y
0	1
1	0

Expression

$$Y = A \text{ NOR } A$$

Experiment No-07

Problem Definition

Design and implement a **1-bit Full Adder** using logic gates. Verify the correctness of the circuit by comparing the generated outputs with the theoretical truth table.

Circuit Diagram

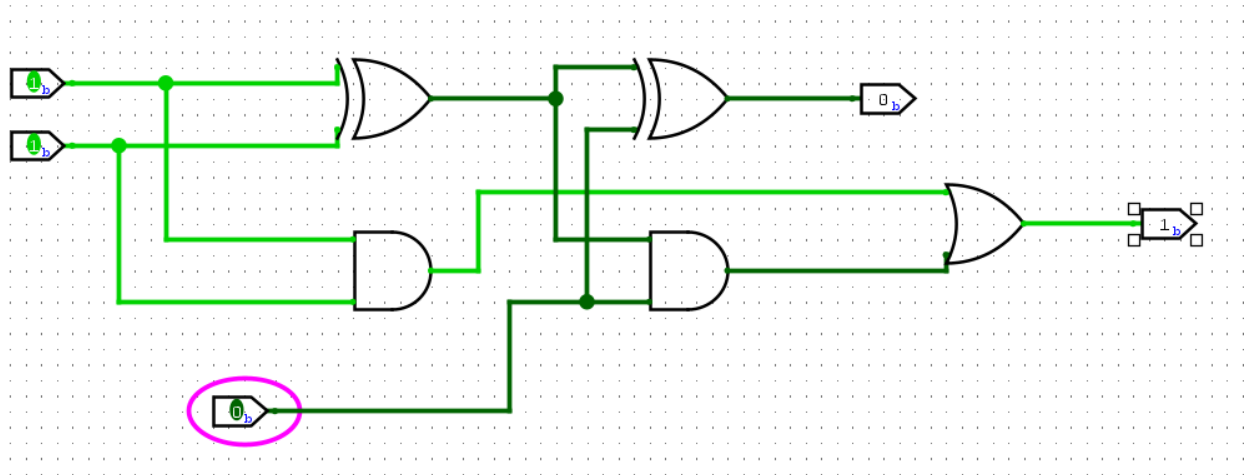


Figure 7: Logic Diagram of 1-Bit Full Adder

Truth Table

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Expressions

$$S = A \oplus B \oplus C_{in}$$

Binary $\rightarrow$ BCD

